

CMT 451 – Mobile Security



Part 1: Emerging Mobile Threats (40 Minutes)

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- **IoT-mobile integration risks:** Explain how IoT increases mobile risk exposure by expanding the attack surface.

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- *Practical Example:* Discuss the risks of an executive using their primary smartphone to control smart building functions (like HVAC or automated doors in a modern steel-and-glass eco-facility), creating a bridge from unpatched IoT devices to sensitive corporate email.

- **Advanced mobile malware:** Discuss threats that bypass traditional Mobile Device Management (MDM) solutions. Explain why legacy mobile controls are no longer sufficient.

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- **AI-driven attacks:** Discuss deepfake voice phishing (vishing) targeted at mobile users and how AI will fundamentally change mobile security threats.

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Part 2: Secure-by-Design Principles (40 Minutes)

- **Security embedded at design phase:** Explain why security must start at the design stage rather than being bolted on later. Discuss secure coding practices for mobile platforms.

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- **Privacy-by-design controls:** Detail how to implement privacy-by-design in mobile apps, such as minimizing data collection and localizing processing.

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- **Defense-in-depth for mobile apps:** Map this to the NIST SSDF and ISO/IEC 27001 Secure Operations standards.

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Part 3: Future Trends & Architecture (40 Minutes)

- **Zero Trust mobile access:** Explain the mechanics of Zero Trust in mobile environments, moving away from perimeter VPNs to continuous identity and device posture verification.

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- **AI-enhanced mobile defense:** Explore behavioral biometrics (how the user types, holds the phone) as an authentication layer.

- **Regulatory evolution:** Explain the regulatory impact on mobile design and the future challenges facing mobile security architects

Module 6: Emerging Trends & Secure-by-Design Mobile Systems

Duration: 2 Hours

Standards Alignment: NIST SSDF, ISO/IEC 27001 Secure Operations

1. One-Page Infographic Summary (Textual Template)

- Mobile + IoT Convergence
- Privacy-by-Design
- Secure Application Architecture
- Emerging Mobile Threats
- Future of Mobile Security

2. Detailed Lecture Notes

Emerging Mobile Threats

- IoT-mobile integration risks
- Advanced mobile malware
- AI-driven attacks

Secure-by-Design Principles

- Security embedded at design phase
- Privacy-by-design controls
- Defense-in-depth for mobile apps

Future Trends

- Zero Trust mobile access
- AI-enhanced mobile defense
- Regulatory evolution

3. CAT-Style Scenario Questions

1. Explain why mobile security must start at design stage.
2. How does IoT increase mobile risk exposure?
3. Describe privacy-by-design in mobile apps.

4. A new mobile malware bypasses MDM. What failed?
5. Explain Zero Trust in mobile environments.
6. How will AI change mobile security?
7. Describe secure coding for mobile platforms.
8. Explain regulatory impact on mobile design.
9. Why are legacy mobile controls insufficient?
10. Explain the future challenges of mobile security.